Polarized associated-Higgs production

FCC-ue and muon-collider configurations • Preliminary leading-order results

1. FCC-ue: 5 TeV μ^+ on 183 GeV e^-

Beam configuration: $E_{\mu^+} = 5\,\text{TeV},\ E_{e^-} = 183\,\text{GeV},\ \sqrt{s} \simeq 1.91\,\text{TeV}.$

Polarization: $P_{\mu^+} = +80\%, \quad P_{e^-} = 0.$

Process:

$$\mu^+ e^- \longrightarrow \mu^+ e^- b\bar{b}H \longrightarrow \mu^+ e^- b\bar{b}b\bar{b}.$$

Before four-bottom fiducial cuts	**0.101456** fb
After four-bottom fiducial cuts	**0.081843 ± 0.000255** fb

2. Muon collider: 1 TeV μ^+ on 1 TeV μ^-

Beam configuration: $E_{\mu^+} = 1\,\text{TeV},\ E_{\mu^-} = 1\,\text{TeV},\ \sqrt{s} = 2\,\text{TeV}.$

Polarization: $P_{\mu^+} = +80\%, \quad P_{\mu^-} = 0.$

Process:

$$\mu^+ \mu^- \longrightarrow \mu^+ \mu^- b\bar{b}H \longrightarrow \mu^+ \mu^- b\bar{b}b\bar{b}.$$

Before four-bottom fiducial cuts	**0.116070** fb
After four-bottom fiducial cuts	**0.094913 ± 0.000570** fb

Common fiducial selection

- Four bottom quarks with $|\eta_b| < 5$.
- Two disjoint $b\bar{b}$ invariant-mass windows: $110 < m^{(1)}_{b\bar{b}}, m^{(2)}_{b\bar{b}} < 130\,\text{GeV}$.
- **No minimum bottom-quark transverse-momentum cut.**
- Outgoing charged leptons: $p_T(\ell) > 1\,\text{GeV}$ at generator level.

Notes. Cross sections before the four-bottom fiducial cuts include one model-card branching fraction $\text{BR}(H \to b\bar{b}) = 0.844640$, and the charged-lepton generator-level p_T requirement remains applied. The stated uncertainties are approximate Monte Carlo uncertainties only. Both results are preliminary; the muon-collider run produced 6,539 of the 30,000 requested events and requires further convergence validation.